DESIGN TASK

DESIGN A SPECTRUM SHARING MODEL BETWEEN LTE AND 5G USERS

CO3 – Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems

BL6 – Create / Design | Assessment Weightage: 35%

Academic Information	Details
Course Outcome	CO3
Bloom's Level	BL6 – Create
Assessment	Design / Model based on requirements
Topic	Spectrum Sharing Model Between LTE and 5G Users
Domain	5G/6G Wireless Communication and Radio Resource Management
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Course	5G/6G C ommunication

Design Challenge

Design a dynamic spectrum sharing framework that allows LTE and 5G users to coexist in a common or coordinated spectrum pool. The model must improve spectrum utilization while protecting active LTE users from harmful interference.

Design Vision

The proposed model continuously observes spectrum occupancy, traffic demand, channel quality and interference, then allocates time-frequency resources using priority, QoS and fairness rules.

RUBRICS FOR CALCULATION:

Design Task					
Criteria	**Weightag e**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understandin g of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communicatio n	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communicati on and minimal participation

1. INTRODUCTION AND PROBLEM DEFINITION

1.1 Background

LTE networks already occupy valuable licensed spectrum, while 5G networks require additional capacity for enhanced mobile broadband, ultra-reliable low-latency communication and massive machine-type communication. Static spectrum allocation can leave resources unused during periods of low LTE traffic.

Spectrum sharing provides a way to exploit these underutilized resources. The challenge is to permit 5G access without degrading LTE performance. A coordinated sharing controller can make decisions using occupancy, channel state, traffic demand and interference information.

1.2 Problem Statement

Develop a spectrum sharing model in which LTE and 5G users dynamically share radio resources. The system should identify available resources, estimate interference, allocate resources according to service priority and reclaim resources whenever LTE demand increases.

1.3 Objectives

- Increase spectrum utilization compared with fixed partitioning.
- Protect LTE users using interference and minimum-resource constraints.
- Improve 5G throughput when LTE resources are underutilized.
- Support QoS-aware allocation for eMBB, URLLC and mMTC traffic.
- Maintain fairness between LTE and 5G users.
- Adapt resource allocation to changing traffic and channel conditions.
- Provide a scalable model for multi-cell and future 6G deployment.

1.4 Key Design Questions

1. How is LTE spectrum occupancy detected?
2. When can a 5G user access LTE resources?
3. How should time-frequency resources be divided?
4. How can interference be limited?
5. How should fairness and priority be handled?
6. How quickly should the system react to LTE traffic changes?

2. SYSTEM REQUIREMENTS AND DESIGN SPECIFICATIONS

2.1 Functional Requirements

- Measure or estimate LTE resource-block occupancy.
- Collect 5G traffic demand and channel-quality information.
- Identify unused or lightly loaded LTE resources.
- Estimate expected LTE–5G interference.
- Allocate resources using priority and fairness.
- Reclaim shared resources when LTE load rises.
- Monitor throughput, latency, SINR, packet loss and utilization.
- Maintain protected resources for important LTE services.

2.2 Non-Functional Requirements

Parameter	Target / Goal	Purpose
Spectrum utilization	Higher than static partition	Reduce unused capacity
LTE protection	Interference below threshold	Protect legacy users
5G latency	Low for priority traffic	Support URLLC-type services
Fairness	High Jain index	Avoid starvation
Allocation	Dynamic each scheduling interval	Adapt to traffic
Scalability	Multiple cells/users	Large networks
Security	Authenticated control	Prevent malicious decisions

2.3 Traffic Classes

Class	Example	Priority
LTE critical	Voice/control	High
LTE broadband	Data users	Medium
5G URLLC	Industrial control	Very high
5G eMBB	Video/high-rate data	High
5G mMTC	IoT devices	Medium/low

2.4 Design Assumptions

- LTE and 5G base stations can exchange coordination information.
- A spectrum controller receives occupancy and channel measurements.
- A common or coordinated spectrum pool is available.
- The scheduler can change allocations periodically.
- Safety-critical services retain protected resources.

3. PROPOSED SPECTRUM SHARING ARCHITECTURE

3.1 Layered Architecture

Layer	Elements	Function
User	LTE UEs, 5G UEs	Generate traffic
Radio	LTE eNB, 5G gNB	Radio transmission
Measurement	Sensing, CQI/CSI	Occupancy/channel estimation
Controller	Spectrum sharing controller	Sharing decisions
Scheduler	LTE/5G schedulers	Resource-block allocation
Analytics	SON/core/analytics	Optimization and reporting

3.2 Proposed Flow

LTE / 5G USERS
↓
LTE eNB + 5G gNB
↓
SPECTRUM MEASUREMENT
↓
OCCUPANCY + CSI + INTERFERENCE
↓
SPECTRUM SHARING CONTROLLER
↓
QoS + PRIORITY + FAIRNESS SCHEDULER
↓
TIME–FREQUENCY RESOURCE ALLOCATION
↓
LTE + 5G TRANSMISSION

3.3 Centralized Model

A centralized controller collects information from LTE and 5G base stations and computes a common allocation. It is easy to simulate and provides a global view of spectrum usage.

3.4 Distributed Model

A distributed approach allows neighboring base stations to coordinate locally. It reduces dependence on one controller but requires reliable exchange of measurements and conflict-resolution rules.

3.5 Recommended Academic Design

A centralized controller is recommended for the prototype because the complete decision process can be implemented and evaluated using MATLAB, Python or NS-3. Distributed coordination can be included as future work.

4. SPECTRUM SENSING AND OCCUPANCY ESTIMATION

4.1 Spectrum Sensing

The sharing controller first determines whether LTE resources are occupied, partially occupied, lightly loaded or idle. Measurements can come from received energy, reference signals, scheduling information and channel-state information.

State	Meaning	5G Decision
Occupied	High LTE activity	Do not share
Partially occupied	Some RBs unused	Share approved RBs
Lightly loaded	Low LTE load	Controlled sharing
Idle	No active LTE traffic	Allow sharing subject to policy
Protected	Reserved LTE service	Block sharing

4.2 Sensing Methods

- Energy detection using an adaptive threshold.
- Cyclostationary detection for signal identification.
- Reference-signal-based LTE occupancy estimation.
- Network-assisted sensing using eNB scheduling information.
- CQI/CSI-based channel quality and interference estimation.

4.3 Energy Detection Model

$$\mathbf{E = (1/N)\ \Sigma\ |x[n]|^2}$$

If measured energy E is above threshold λ, the resource can be classified as occupied. If E is below λ, it is considered a candidate for sharing. In practice, the threshold should adapt to noise and interference.

4.4 Sensing Challenges

- Noise uncertainty can create false detections.
- Hidden transmitters may not be observed by all nodes.
- Fading changes measured signal strength.
- Adjacent-channel interference can affect sensing.
- Frequent sensing adds signaling and processing overhead.

5. RESOURCE ALLOCATION AND SHARING ALGORITHM

5.1 Resource Representation

The common spectrum is represented as a time-frequency grid. Every resource block is marked as LTE-reserved, 5G-available, conditionally shared or blocked.

5.2 Allocation Inputs

- LTE occupancy and load.
- 5G traffic demand.
- CQI/CSI.
- Estimated interference.
- User/service priority.
- Queue length.
- Historical utilization.
- Fairness state.

5.3 Allocation Score

$$\mathbf{Score = w_1 \cdot Availability + w_2 \cdot CQI + w_3 \cdot Priority + w_4 \cdot Demand - w_5 \cdot Interference}$$

Higher scores indicate better candidate resources. Before allocation, any resource violating the LTE interference constraint is removed.

5.4 Proposed Algorithm

1. Collect LTE occupancy, CSI/CQI and traffic information.

2. Collect 5G demand and QoS requirements.

3. Identify candidate unused/lightly loaded RBs.

4. Estimate interference for each candidate.

5. Reject unsafe RBs.

6. Calculate allocation scores.

7. Allocate high-priority traffic first.

8. Apply fairness constraints.

9. Reclaim resources when LTE load increases.

10. Record performance metrics and repeat.

6. INTERFERENCE MANAGEMENT AND FAIRNESS

6.1 Main Interference Sources

- Co-channel LTE–5G interference.
- Adjacent-channel leakage.
- Inter-cell interference.
- High-power 5G transmission near LTE receivers.
- Dense-user contention.

6.2 Protection Constraint

$$I_{5\to L} \le I\_threshold$$

A 5G transmission is permitted on a shared resource only when estimated interference at the protected LTE receiver remains below the configured limit.

6.3 Mitigation Techniques

Technique	Action	Benefit
Power control	Reduce 5G power on shared RBs	Lower interference
Guard resources	Reserve sensitive RBs	Better protection
Beamforming	Direct energy to intended users	Spatial isolation
Resource separation	Use different RBs/time slots	Avoid collisions
QoS scheduling	Prioritize critical LTE	Protect important traffic
Dynamic reassignment	Move 5G when LTE load rises	Adaptation

6.4 Fairness

$$J = (\Sigma x_i)^2 / [n\ \Sigma x_i^2]$$

Jain's fairness index can evaluate whether LTE and 5G users receive balanced service. A value close to 1 indicates high fairness.

6.5 Protection Policy

- Maintain minimum LTE resource reservation.
- Permit 5G only on approved resources.
- Reduce sharing when LTE load crosses a threshold.
- Reclaim shared resources for urgent LTE traffic.
- Move 5G users if interference exceeds the limit.

7. LTE AND 5G MULTIPLE ACCESS DESIGN

7.1 LTE Resource Structure

LTE downlink uses OFDMA, dividing the available bandwidth into orthogonal subcarriers and resource blocks. These resource blocks are convenient units for spectrum-sharing decisions.

7.2 5G NR Resource Structure

5G NR uses flexible OFDM-based transmission and supports different numerologies and slot configurations. The proposed model therefore coordinates time-frequency resources rather than assuming that all resources are permanently assigned to one generation.

7.3 Multiple Access Strategy

Strategy	Role in Model
OFDMA	Divide spectrum into schedulable resource blocks
Frequency-domain sharing	Assign different RB groups to LTE/5G
Time-domain sharing	Alternate transmissions in scheduling intervals
QoS scheduling	Prioritize URLLC/eMBB/LTE services
Spatial reuse	Use MIMO/beamforming for additional separation

7.4 Example Time-Frequency Map

FREQUENCY →

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| LTE | LTE | 5G  | 5G  |
| LTE | 5G  | LTE | 5G  |
```

TIME →

7.5 Spectrum Reclamation

When LTE traffic increases, the controller reclaims shared resource blocks. The 5G scheduler receives the updated resource map and moves affected users to remaining resources.

8. PERFORMANCE ANALYSIS AND EVALUATION

8.1 Performance Metrics

Metric	Measurement	Goal
Spectrum utilization	Used RBs / total RBs	Higher than static allocation
LTE throughput	LTE received bits/time	Maintain minimum target
5G throughput	5G received bits/time	Increase during idle LTE periods
SINR	Signal/(interference+noise)	Above service threshold
Interference	Estimated/received power	Below protection limit
Latency	Receive time − transmit time	Low for priority users
Fairness	Jain index	Close to 1
Blocking probability	Rejected/total requests	Low

8.2 Baseline Comparison

Model	LTE Protection	5G Utilization	Efficiency
Static partition	High	Limited	Medium
LTE-only	Very high	None	Low during idle periods
Uncontrolled sharing	Low	High	Unstable
Proposed model	High	High	High

8.3 Test Scenarios

1. Low LTE traffic with increasing 5G demand.
2. High LTE traffic with low 5G demand.
3. Peak LTE and 5G load.
4. Sudden LTE traffic burst during 5G sharing.
5. High interference condition.
6. Different CQI values among 5G users.
7. Multi-cell simultaneous sharing.
8. Controller failure and recovery.

8.4 Expected Trend

When LTE traffic is low, more LTE resources can be reused by 5G, increasing overall utilization and 5G throughput. As LTE load rises, the controller automatically reduces 5G access to protect LTE service.

9. SIMULATION / PROTOTYPE IMPLEMENTATION

9.1 Recommended Tools

Tool	Purpose
MATLAB	Allocation algorithm and graphs
Simulink	System-level communication model
Python	Optimization and numerical evaluation
NS-3	Packet-level network simulation
5G Toolbox	5G waveform/link-level studies
Excel	Result tables and comparison

9.2 Simulation Flow

LTE + 5G TRAFFIC GENERATION

↓

COMMON TIME-FREQUENCY GRID

↓

LTE OCCUPANCY ESTIMATION

↓

CQI/CSI + INTERFERENCE ESTIMATION

↓

SPECTRUM SHARING CONTROLLER

↓

RESOURCE ALLOCATION

↓

SINR / THROUGHPUT / LATENCY / FAIRNESS

↓

COMPARE WITH STATIC PARTITION

9.3 Example Parameters

Parameter	Example Value
Shared bandwidth	20 MHz
LTE users	10–20
5G users	10–20
Scheduling interval	1 ms or selected interval
LTE minimum reservation	30–50% depending on scenario
SNR sweep	0–30 dB
Simulation duration	10–100 s

9.4 Implementation Steps

1. Generate LTE and 5G traffic.

2. Create common resource grid.

3. Assign LTE reservations.

4. Generate changing traffic loads.

5. Calculate resource availability.

6. Apply interference/fairness constraints.

7. Run proposed scheduler.

8. Calculate metrics.

9. Compare with static partitioning.

10. Plot results.

10. OPTIMIZATION, SECURITY AND SCALABILITY

10.1 Optimization Objective

The sharing controller can maximize a weighted utility containing LTE throughput, 5G throughput, spectrum utilization and fairness while satisfying LTE protection constraints.

$$\textbf{Maximize } U = \alpha T\_LTE + \beta T\_5G + \gamma \text{Utilization} + \delta \text{Fairness}$$

Subject to: LTE SINR ≥ SINR_min, interference ≤ threshold, minimum LTE reservation and available-resource constraints.

10.2 Optimization Methods

• Weighted proportional fairness.

• Greedy resource-block allocation.

• Linear/integer optimization for small systems.

• Genetic algorithms for complex scenarios.

• Reinforcement learning for adaptive sharing.

• Multi-agent learning for distributed cells.

10.3 Security Threats

Threat	Impact	Protection
False spectrum reports	Wrong allocation	Authenticated measurements
Unauthorized access	Interference	Identity verification
Controller attack	Large-scale disruption	Secure control channel
Data manipulation	Incorrect decisions	Integrity protection
Jamming	Reduced availability	Detection and adaptive reassignment

10.4 Scalability

For large networks, a hierarchical architecture can use cell-level controllers for fast decisions and a higher-level coordinator for inter-cell policy. This reduces controller overload and supports many cells.

10.5 Future 6G Extension

- AI-native spectrum management.
- Reconfigurable intelligent surfaces.
- Cell-free massive MIMO.
- Integrated sensing and communication.
- Sub-THz spectrum sharing.
- Digital-twin-based optimization.

11. CO3 MAPPING, EXPECTED RESULTS AND CONCLUSION

11.1 CO3 Mapping

CO3 Requirement	Implementation
Spectrum access strategies	Dynamic identification and allocation of underutilized LTE resources
Efficient wireless transmission	CQI-aware allocation and interference management
Multiple access techniques	OFDMA-based time-frequency resource-block scheduling
5G communication	5G users coexist with LTE using coordinated allocation
BL6 design	Complete model with requirements, algorithm, constraints and evaluation

11.2 Expected Results

- Higher spectrum utilization than fixed partitioning.
- Improved 5G throughput during low LTE load.
- Protection of LTE through interference thresholds.
- Improved fairness between generations.
- Dynamic response to traffic changes.
- Reduced resource wastage.
- Clear extension path toward 6G.

11.3 Applications

- Private 5G networks deployed alongside LTE.
- Smart factories migrating gradually from LTE to 5G.
- Enterprise and campus networks.
- Industrial IoT systems.
- Mission-critical wireless services.
- Dynamic spectrum access environments.

11.4 Final Conclusion

The proposed LTE–5G spectrum sharing model provides a controlled and adaptive method for improving radio-resource utilization. It combines spectrum sensing, occupancy estimation, channel-quality information, interference constraints, QoS priority and fairness in a single scheduling framework. The design directly demonstrates CO3 concepts related to spectrum access and multiple access techniques.

The model can be implemented in MATLAB, Simulink, Python or NS-3 and compared against static LTE/5G partitioning. Future improvements can introduce machine-learning-based resource allocation, multi-cell coordination, advanced beamforming and 6G spectrum-management techniques.

References / Standards to Consult

- 3GPP specifications for LTE/E-UTRA and 5G NR.
- 3GPP radio resource management and QoS concepts.
- ETSI/3GPP studies on spectrum sharing and coexistence.
- ITU-R recommendations related to radio-spectrum management.
- IEEE references on OFDM/OFDMA and wireless resource allocation.